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Jeff Shen

Education

H.B.Sc., University of Toronto 2018-2022
Astronomy & Astrophysics, Statistics, Mathematics. Dean's List Scholar.

Publications

2. **Jeff Shen**, Gwendolyn M. Eadie, Norman Murray, Dennis Zaritsky, Joshua S. Speagle, Yuan-Sen Ting, Charlie Conroy, Phillip A. Cargile, Benjamin D. Johnson, Rohan P. Naidu, Jiwon Jesse Han. The Mass of the Milky Way from the H3 Survey, submitted to ApJ.

1. **Jeff Shen**, Allison W. S. Man, Johannes Zabl, Zhi-Yu Zhang, Mikkel Stockmann, Gabriel Brammer, Katherine E. Whitaker, and Johan Richard. Molecular gas in a gravitationally lensed galaxy group at $z = 2.9$. 2021, ApJ, 917, 79. 2105.11572.

Research Experience

Undergraduate Research Course 2021 Sep - Present
Department of Astronomy and Astrophysics, University of Toronto
Supervisor: Prof. Roberto Abraham
Software for the Dragonfly Telephoto Array.

Independent Research 2021 Sep - Present
Supervisors: Prof. Gwendolyn Eadie, Prof. Alex Stringer
Diagnostics for non-iterative Bayesian approximations.

Independent Research 2021 Jul - Present
Supervisor: Prof. Norman Murray
Using Stan to fit dynamical models for the angular momentum of the Earth-Moon system to constrain the strength of thermal tides over geological times.

Independent Research 2021 May - Present
Supervisor: Dr. Joshua Speagle
Probabilistic predictions of stellar ages with normalizing flows. Quantifying the contribution of stellar and Galactic information to age estimates.

Undergraduate Summer Research Fellowship 2021 May - 2021 Aug
Canadian Institute for Theoretical Astrophysics
Supervisors: Prof. Gwendolyn Eadie, Prof. Norman Murray, Dr. Joshua Speagle
Continued work on the mass distribution of the Milky Way. Collaborated with members of the H3 Survey and submitted a first-author paper to ApJ.

Undergraduate Research Course 2020 Aug - 2021 May
Canadian Institute for Theoretical Astrophysics
Supervisors: Prof. Gwendolyn Eadie, Prof. Norman Murray
Developed a hierarchical Bayesian model for modelling the mass of galactic halos using a fast and scalable sampling algorithm.

Summer Undergraduate Research Program 2020 May - 2020 Aug
Dunlap Institute for Astronomy and Astrophysics, University of Toronto
Supervisor: Prof. Allison Man
Analyzed ALMA data to infer molecular gas masses in several gravitationally lensed high-redshift galaxies. Presented results in a first-author publication in ApJ.

Presentations

SDSS 2021 Collaboration Meeting, JHU/Online 2021 August
Lightning talk: "Predicting the ages of stars with machine learning."
Stellar Stats Workshop, UofT/Online 2021 May
Lightning talk: "Estimating the mass distribution of the Milky Way with Bayesian multilevel models."
Canadian Astronomical Society (CASCA) AGM, NRC Herzberg/Online 2021 May
Poster presentation and lightning talk: "Molecular gas in a gravitationally lensed galaxy group at z = 2.9."
Vancouver Area Cosmology Meeting, UBC/SFU/TRIUMF/Online 2021 Apr
Talk: "Molecular gas in gravitationally lensed galaxies at z = 2.9."
REQUIEM Galaxies Telecon, Online 2021 Oct
Talk: "Molecular gas in gravitationally lensed galaxies at z = 2.9."
Summer Undergraduate Research Program Poster Session, UofT/Online 2021 Aug
Poster presentation: "Molecular gas in a gravitationally lensed galaxy group at z = 2.9." Awarded prize for one of top three posters.

Honours and Awards

Undergraduate Summer Research Fellowship, ($8,396 CAD) 2021 May - 2021 Aug
Summer Undergraduate Research Program Award ($9,500 CAD) 2020 May - 2020 Aug
Dunlap Institute for Astronomy and Astrophysics, University of Toronto

Selected Projects

4. Comprehensive scientific computing package for Rust: linear algebra, statistics, numerical optimization, linear models, and more. `github.com/al-jshen/compute`
3. Zero-dependency reverse-mode automatic differentiation package. `github.com/al-jshen/reverse`
2. Fast and easy random number generation in Rust with Wyrand. `github.com/al-jshen/alea`
1. Parallelized ray tracer for realistic scene rendering. `github.com/al-jshen/traycer`

Media

National Ratio Astronomy Organization (NRAO) eNews 2021 Jul
Digital newsletter feature: "Molecular gas in high redshift galaxies". `https://science.nrao.edu/enews/14.7/index.shtml#molecular_gas`
SURP Student of the Week Feature 2020 Jul
Interview: `https://www.dunlap.utoronto.ca/surp-sow/sow12020/`

Outreach

Space Science Journalist (Current in Space segment), *The Star Spot* podcast 2019 - 2021
Curate, write and record stories about research and news in astronomy. Episodes 177 - end of show.

Volunteering

Python Workshop Lead, Sir Winston Churchill Secondary, Vancouver, B.C. 2015 - 2018
Coordinated and delivered workshops to prepare students for national programming contest.

Technical Skills

Data: Stan, Pyro, Pytorch, NumPy, SciPy, Pandas, Scikit-Learn, Matplotlib, SQL, R
Other: C, C++, Rust, BLAS/LAPACK, Astropy, LaTeX, regex, bash, git, Docker, AWS, Javascript, React